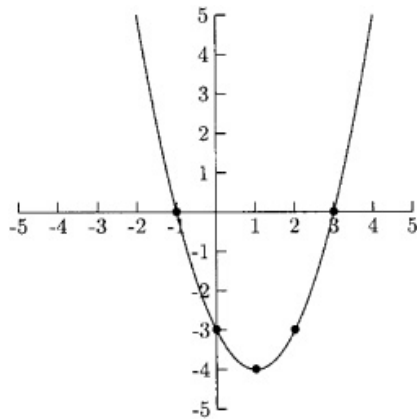


PRELIMINARY MATHEMATICS ASSESSMENT TASK 2 (PRELIMINARY)

SECTION A: Multiple Choice (4 marks)

Question 1

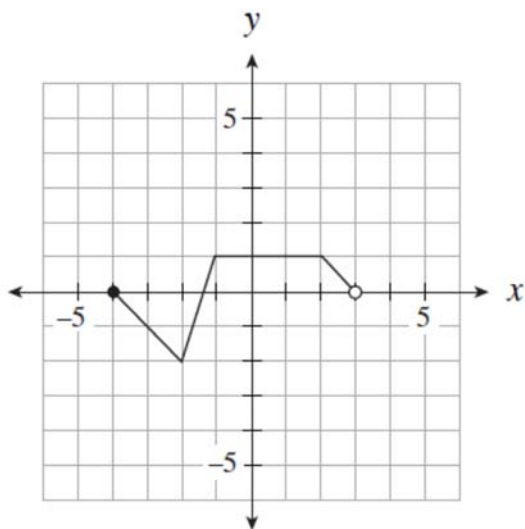
What type of function is shown by the graph below?



- a) Linear
- b) Quadratic
- c) Cubic
- d) Exponential

Question 2

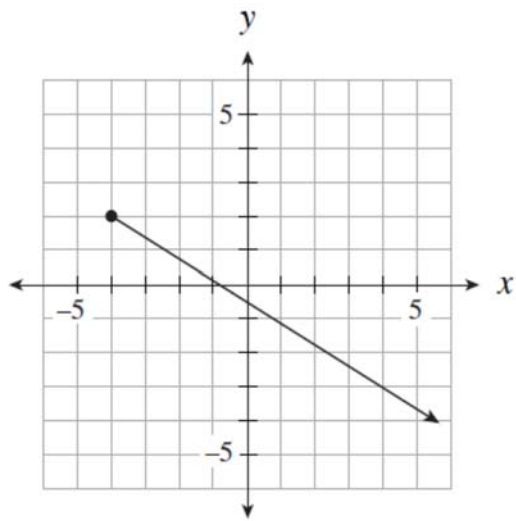
Determine the **domain** of the following relation



- a) $-4 < x \leq 3$
- b) $-4 \leq x \leq 3$
- c) $-2 \leq y \leq 1$
- d) $-4 \leq x < 3$

Question 3

Determine the **range** of the following function



- a) $x \geq -4$
- b) $y \geq -4$
- c) $y \geq 2$
- d) $y \leq 2$

Question 4

Consider the graph of the function $y = 2^{x-1} - 1$. At what point will this graph intersect the x -axis?

- a) $(0, 0)$
- b) Never intersects the x -axis
- c) $(1, 0)$
- d) $(2, 0)$

SECTION B

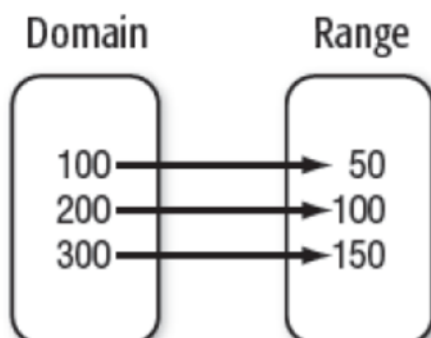
Question 5 (10 marks)

Marks

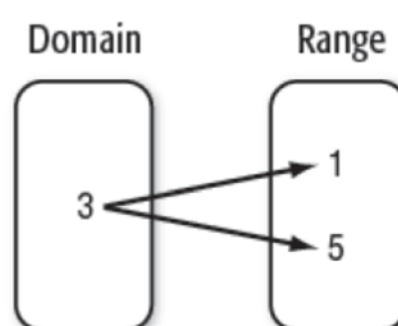
- (a) For the following graphs, determine whether the following relations are functions

4

(i)



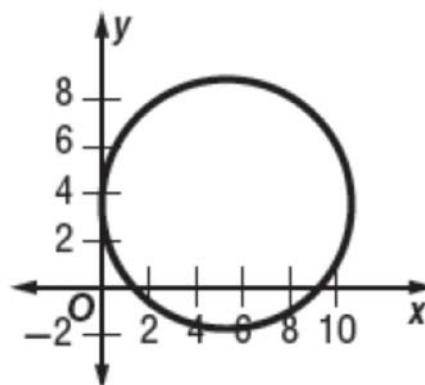
(ii)



(iii)

x	y
1	2
2	4
3	6

(iv)



- (b) Consider the function $f(x) = 3x^2 - x$, and $g(x) = 2 - x$.

(i) Evaluate $f(-1)$

1

(ii) Evaluate $g(b)$

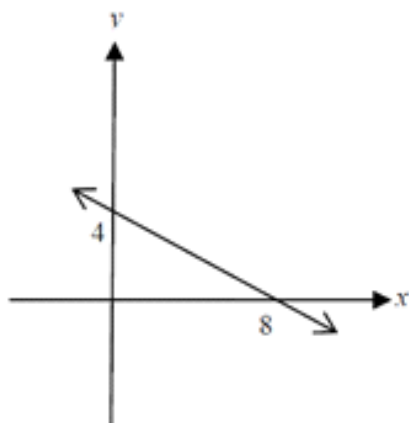
1

(iii) Find a if $f(a + 1) = f(a - 1)$

2

- (c) Write down the equation of the line represented in the diagram:

2



Question 6 (9 marks)**Marks**

On separate diagrams, sketch the graphs of the following, showing **essential features**

- | | | |
|-----|----------------------|---|
| (a) | $3x - 2y + 12 = 0$ | 2 |
| (b) | $y = x^2 - 4x + 3$ | 3 |
| (c) | $y = x + 2 $ | 2 |
| (d) | $y = \sqrt{9 - x^2}$ | 2 |

Question 7 (9 marks)**Marks**

- (a) State the **domain** and **range** of each of the following functions.

- | | | |
|-------|--------------------|---|
| (i) | $y = \sqrt{2 - x}$ | 2 |
| (ii) | $x^2 + 3 = y$ | 2 |
| (iii) | $y = 3^x$ | 2 |
- (b) Given the function $y = \frac{2}{x^2 - 1}$
- | | | |
|------|---|---|
| (i) | Find the vertical asymptote(s) | 2 |
| (ii) | Find the horizontal asymptote(s) | 1 |

Question 8 (9 marks)**Marks**

- (a) State whether the following are ODD, EVEN or NEITHER. Justify your answer.

- | | | |
|-------|----------------------------|---|
| (i) | $f(x) = 1 - x^2$ | 2 |
| (ii) | $f(x) = \frac{x}{1 + x^2}$ | 2 |
| (iii) | $f(x) = -3 x $ | 2 |
- (b) Find the centre and radius of the circle
- $x^2 + y^2 - 4x + 6y - 3 = 0$
- 3

Question 9 (12 marks)**Marks**

- (a) A function $f(x)$ is defined by

$$f(x) = \begin{cases} 0, & x < -2 \\ -1, & -2 < x < 0 \\ x, & x \geq 0 \end{cases}$$

- (i) Find $f(-4) + f\left(-\frac{1}{2}\right) + f(0)$ **2**
- (ii) Sketch the graph of $y = f(x)$ **2**
- (b) If $y = \frac{1}{3-x} + 1$
- Explain how the above function is transformed from the curve $y = \frac{1}{x}$ **3**
- (c) Find the **range** of $f(x) = 3 + 2x - x^2$, $0 \leq x < 5$ **3**
- (d) If $f(x) = x^2 - 4$ and $g(x) = x - 2$, find in simplest form $f(g(x))$ **2**

End of Examination

1. B. 2. D. 3. D 4. C

Q5.

a) i) Function (1) ii) Not Function (1) iii) Function (1) iv) Not Function (1)

b) i) $f(-1) = 3(-1)^2 - (-1)$

$$= 3 + 1$$

$$= 4$$

(1)

ii) $g(b) = 2 - b$

(1)

iii) $f(a+1) = 3(a+1)^2 - (a+1)$

$$= 3(a^2 + 2a + 1) - (a + 1)$$

$$= 3a^2 + 6a + 3 - a - 1$$

$$= 3a^2 + 5a + 2$$

$$f(a-1) = 3(a-1)^2 - (a-1)$$

(1)

$$= 3(a^2 - 2a + 1) - (a - 1)$$

$$= 3a^2 - 6a + 3 - a + 1$$

$$= 3a^2 - 7a + 4$$

$$\therefore f(a+1) = f(a-1)$$

$$3a^2 + 5a + 2 = 3a^2 - 7a + 4$$

$$12a = 2$$

$$a = \frac{1}{6}$$

(1)

c) y int: 4

x int: 8.

$y = mx + b$ where $b = 4$

(1).

$$\therefore y = mx + 4$$

when $x = 8$ $y = 0$

$$\therefore m = -\frac{1}{2}$$

(1)

$$\therefore y = -\frac{1}{2}x + 4.$$

/ 10

Question 6

a) $3x - 2y + 12 = 0$

x int: $3x + 12 = 0$

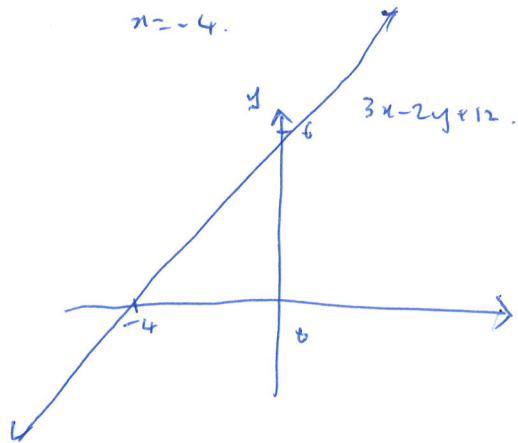
$$3x = -12$$

$$x = -4$$

y int: $-2y + 12 = 0$

$$2y = 12$$

$$y = 6$$



(2)

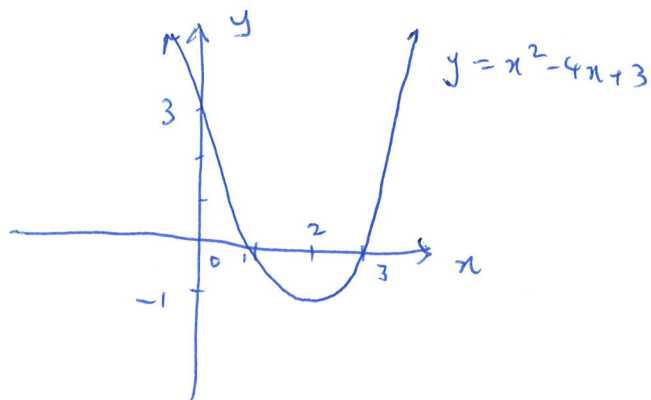
b) $y = x^2 - 4x + 3$

$$= (x - 3)(x - 1)$$

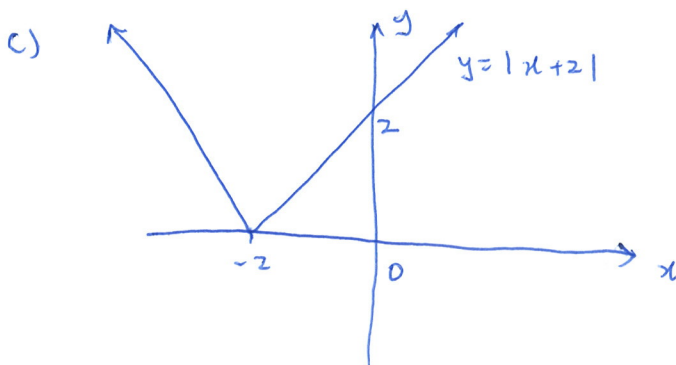
x int: $x = 3, x = 1$

y int: $y = 3$

vertex at $x = 2 \therefore V(2, -1)$

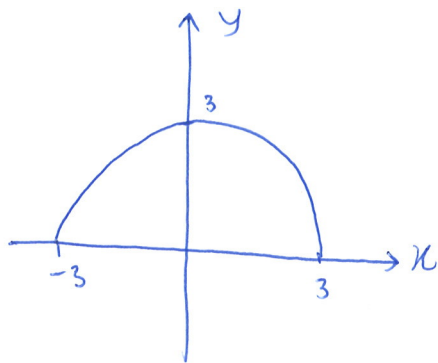


(3)



(2)

d)



✓✓

Question 7.

- a) i) Domain: $2 - x \geq 0$
 $x \leq 2$
 Range: $y \geq 0$

✓

✓

- ii) Domain: All real x
 Range: $y \geq 3$

✓

✓

- iii) Domain: All real x
 Range: $y > 0$

✓

✓

b)
$$y = \frac{2}{(x-1)(x+1)}$$

- i) vertical asymptote at
 $x = 1, x = -1$

✓✓

- ii) horizontal asymptote
 as $x \rightarrow \infty, y \rightarrow 0$.

~~as~~ as $x \rightarrow \infty$ $y = 0$ is horizontal asymptote

✓

Question 8

a) i) $f(x) = 1 - x^2$

$$f(-x) = 1 - (-x)^2 \quad (\checkmark)$$

$$= 1 - x^2$$

$$= f(x)$$

$\therefore$ even (✓)

ii) $f(x) = \frac{x}{1+x^2}$

$$f(-x) = \frac{-x}{1+(-x)^2} \quad (\checkmark)$$

$$= -\frac{x}{1+x^2}$$

$$= -f(x)$$

$\therefore$ odd. (✓)

iii) $f(x) = -3|x|$

$$f(-x) = -3|-x|$$

$$= -3|x|$$

$$= -f(x) \quad (\checkmark)$$

$$= \text{even} \quad (\checkmark)$$

b) $x^2 + y^2 - 4x + 6y - 3 = 0$

$$x^2 - 4x + y^2 + 6y = 3$$

$$(x-2)^2 + (y+3)^2 = 3 + 4 + 9. \quad (\checkmark)$$

$$(x-2)^2 + (y+3)^2 = 16. \quad (\checkmark)$$

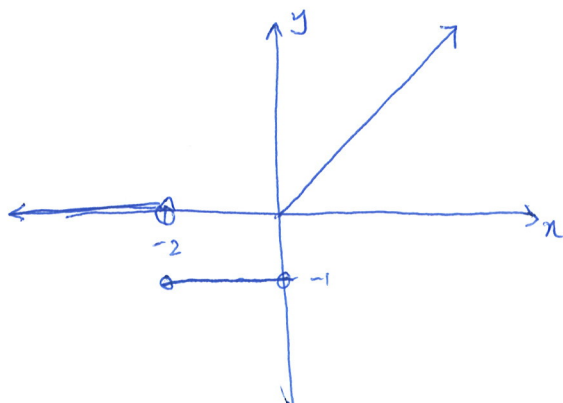
$\therefore$ Centre $(2, -3)$ radius 4. (✓)

Question 9

a) i) $f(-4) = 0$
 $f(-\frac{1}{2}) = -1$
 $f(0) = 0$

$\therefore f(-4) + f(-\frac{1}{2}) + f(0) = 0 - 1 + 0 = -1$

ii)



b) $y = \frac{1}{3-x} + 1$
 $= \frac{1}{-(x-3)} + 1$

$y = \frac{1}{-x} \rightarrow y = \frac{1}{-x}$
 $\rightarrow y = \frac{1}{-(x-3)}$
 $y = \frac{1}{-(x-3)} + 1$

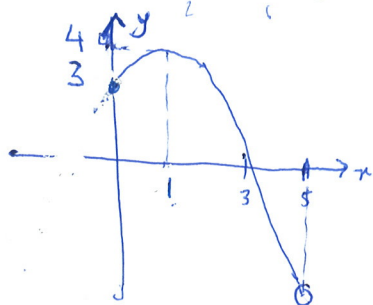
reflect about x/y -axis.

shift right by 3 units

move up by 1 unit.

c) $f(x) = -(x^2 - 2x - 3)$

$= -(x-3)(x+1)$



-1

range -12

$f(x) \leq 4$

d)

$f(x) = (x-2)^2 - 4$

$= x^2 - 4x + 4 - 4$

$= x^2 - 4x$

$= x(x-4)$